Cross-Architecture Contrastive Learning for Few-Shot Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification is a fundamental task in remote sensing with broad applications in precision agriculture, environmental monitoring, and related fields. However, existing models encounter significant difficulties when labeled samples are scarce. Although contrastive learning has mitigated the dependence on labeled data, most current methods remain limited to a single network architecture, thus failing to exploit the complementary strengths among heterogeneous representations. To address this issue, we propose a cross-architecture contrastive learning (CACL) framework for few-shot HSI classification, which integrates convolutional neural networks (CNNs) and graph convolutional networks (GCNs) through a parallel dual-branch design. The CNN branch captures fine-grained local spectral–spatial representations, while the GCN branch models global graph-structural dependencies. Positive sample pairs are constructed by aligning these heterogeneous features, and their consistency is maximized within a shared embedding space to achieve deep cross-architecture fusion. Moreover, a spectral frequency-domain enhancement module (SFEM) is introduced into the CNN branch to refine local features via frequency-domain analysis, and an adaptive multiscale graph convolution module (AMGCM) is incorporated into the GCN branch to capture richer hierarchical spatial features. Experiments on the WHU-Hi-HanChuan (WHU), Indian Pines (IP), and University of Pavia (UP) datasets demonstrate that our method achieves overall classification accuracies of 95.29%, 96.73%, and 99.17%, respectively, with extremely limited training samples, significantly outperforming multiple mainstream baseline models.**

***Index Terms*—Contrastive learning (CL), convolutional neural network (CNN), cross-architecture (CA), deep learning (DL), graph convolutional network (GCN), hyperspectral image (HSI) classification, remote sensing.**

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I. Introduction

HYPERSPECTRAL imaging is an advanced remote sensing technology that captures hundreds of narrow, contiguous spectral bands to provide rich spectral information, thereby precisely revealing the physical and chemical properties of land cover [1], [2], [3], [4], [5], [6], [7], [8]. As a core task in remote sensing, hyperspectral image (HSI) classification aims to assign each pixel in an image to a specific category [9], [10], [11], [12]. It has been widely applied in precision agriculture, environmental monitoring, urban planning, and mineral exploration.

Early HSI classification methods based on traditional machine learning, such as support vector machines [13], random forests [14], and k-nearest neighbors [15], though theoretically mature, rely on manually engineered features and shallow models. This makes them ill-suited to handle the high dimensionality and complex nonlinear structures of the data, resulting in limited performance. The pursuit of higher classification performance has driven a paradigm shift from traditional machine learning to deep learning. The powerful representational learning capability of deep learning provides a key breakthrough for addressing the feature extraction challenges in HSIs [16], [17], [18], [19], [20]. Among various deep learning models, convolutional neural networks (CNNs) have rapidly become the mainstream framework for HSI classification due to their strong feature extraction ability. This evolution ranges from 1D-CNN [21] that focuses solely on the spectral dimension, to 2D-CNN [22] capable of capturing spatial contextual information simultaneously, and then to 3D-CNN [23], [24], [25], [26], specifically designed for hyperspectral data cubes that enables integrated learning of joint spatial–spectral features, often enhanced with residual connections [27], [28], [29] or attention mechanisms [30]. Overall, the core advantage of CNNs lies in their ability to effectively extract translation-invariant local features through local convolution and weight sharing. However, CNN-based methods also have obvious limitations: they struggle to capture global features effectively. Owing to their localized receptive fields and fixed convolutional kernels, CNNs find it difficult to model semantic relationships among distant pixels, thereby

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constraining their ability to represent complex land-cover distributions comprehensively. To address this, Transformer-based [31], [32], [33] architectures have been explored for capturing long-range dependencies via self-attention mechanisms. However, these models typically demand extensive labeled data to establish global correlations, limiting their direct applicability in few-shot scenarios.

In contrast, graph convolutional networks (GCNs) have been introduced into HSI classification due to their proficiency in learning global features [34], [35], [36], [37], [38]. GCNs represent data as graph structures, aggregating neighborhood information through message-passing mechanisms, and are inherently adept at modeling global topological relationships. Early approaches involved constructing computationally expensive pixel-level graphs [34], while later methods adopted superpixel segmentation to generate superpixel-level graphs for improved efficiency [39]. Unlike CNNs with fixed local receptive fields, GCNs can directly capture complex relationships between nodes at any distance, effectively solving the challenge of modeling long-range dependencies. However, GCNs also have inherent limitations: superpixel-level processing often loses fine-grained pixel-level information, resulting in weaker discriminative ability for local structures such as object boundaries and textures.

Thus, CNNs and GCNs exhibit natural complementarity in feature extraction—CNNs are adept at capturing local details, while GCNs excel at modeling global structures. Existing approaches like WFCG [40] combine pixel-level CNNs with superpixel-level graph attention networks via a weighted feature fusion mechanism to improve classification performance in few-shot scenarios. NESSGGCN [41] further optimizes classification accuracy by integrating gated graph convolution with CNNs, combining global features from non-Euclidean space with local features from Euclidean space. Recent advancements have further validated the potential of such integration strategies. Notably, Wang et al. demonstrated that multiscale interactive [42] and multistage complementary [43] fusion mechanisms can significantly enhance the robustness of feature representation. Furthermore, their proposed global–local feature coupling network [44] has achieved superior performance by effectively coupling features from CNN and Transformer architectures.

In recent years, contrastive learning-based HSI classification methods have rapidly evolved, effectively mitigating the scarcity of labeled data through self-supervised or semi-supervised paradigms [45], [46], [47], [48], [49]. The core of these approaches lies in utilizing contrastive loss functions to bring positive pairs closer and push negative pairs farther apart, thereby learning discriminative feature representations from unlabeled data. SSCL [50] constructs positive pairs by applying Gaussian noise and leverages unsupervised contrastive learning to improve classification performance. ConGCN [51] achieves high-precision classification with few labeled samples by combining adaptive graph augmentation with semi-supervised contrastive loss. UMGC [52] enables efficient feature extraction by constructing multiview graph structures integrated with unsupervised contrastive learning. Despite significant progress in existing contrastive learning approaches, their core methodologies remain largely confined to feature alignment within a single network architecture, failing to fully exploit the complementary potential of features extracted from different architectures. Specifically, GCN excels at capturing global structural relationships of land cover, while CNNs are better suited for pixel-level local spectral–spatial details. Though semantically consistent, these features exhibit heterogeneous distributions. Existing approaches rarely explicitly align cross-architecture features as positive pairs, hindering the collaborative optimization of local details and global structures within a unified representation space. This limitation constrains models' comprehensive characterization of complex scenes.

To address this, we propose a cross-architecture contrastive learning (CACL) mechanism that innovatively pairs global graph structural features from the GCN branch with local pixel features from the CNN branch as positive sample pairs. By maximizing their consistency in a shared embedding space, we achieve explicit alignment and deep fusion of heterogeneous features. To further enhance the quality and robustness of positive sample pairs under few-shot conditions, we designed two feature enhancement modules: a spectral frequency-domain enhancement module (SFEM) in the CNN branch, which extracts finer spectral global patterns through frequency-domain analysis; an adaptive multiscale graph convolution module (AMGCM) in the GCN branch, which uncovers richer spatial hierarchical contextual information via multiscale graph construction. The main contributions of this article are summarized as follows.

1) We propose a cross-architecture contrastive learning (CACL) mechanism that innovatively aligns the local detail features of CNN with the global structural features of GCN as positive pairs, achieving deep fusion of heterogeneous features.
2) A spectral frequency-domain enhancement module (SFEM) is designed, which leverages frequency-domain analysis to extract finer spectral patterns under data-limited conditions, thereby augmenting the discriminative capacity of local spectral features.
3) An adaptive multiscale graph convolution module (AMGCM) is introduced, enabling richer spatial hierarchical contextual information to be mined from scarce samples through a multiscale graph construction mechanism with learnable scale weighting, and consequently enhancing the robustness of global spatial representations.

The remainder of this article is organized as follows. Section II presents the proposed CACL method in detail. Section III elaborates on the experimental results obtained on three HSI classification benchmark datasets. Finally, Section IV concludes this article.

II. Method

A. Overall Framework

The flowchart illustrating the proposed method is shown in Fig. 1. The framework tackles the feature learning problem in scenarios with scarce labeled data by synergistically leveraging

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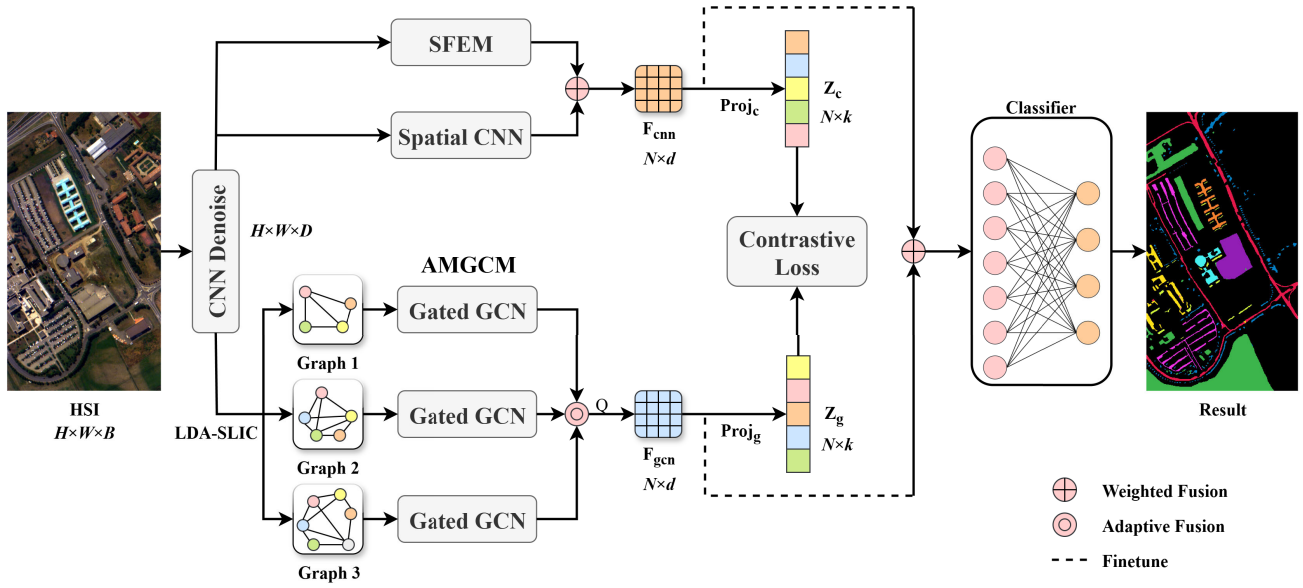


Fig. 1. Framework of the proposed CACL method. It comprises the SFEM and the AMGCM. SFEM extracts discriminative spectral features through frequency-domain transformation, PSD computation, logarithmic compression, and 1-D convolution. AMGCM captures hierarchical spatial features via multiscale graph construction, gated graph convolution, and scale-adaptive fusion. During pre-training, features from the dual branches are projected into a shared embedding space for unsupervised cross-architecture contrastive alignment. In the fine-tuning stage, the classification task is performed through weighted feature fusion and supervised learning.

the advantages of two heterogeneous architectures: CNN and GCN.

The input image is processed in parallel by two branches. The CNN branch incorporates a frequency-domain enhancement mechanism to capture global spectral patterns, which are fused with local spatial features via weighted summation for an enriched representation. The GCN branch, based on an adaptive multiscale graph convolutional feature fusion mechanism, captures hierarchical spatial context through multiscale superpixel graph structures, producing global graph structural features.

The framework introduces CACL, constructing feature pairs extracted from the same pixel by the CNN and GCN branches as positive samples. A two-stage training strategy achieves a balance between feature alignment and classification performance. Through the synergy of three core mechanisms—spectral frequency-domain enhancement, multiscale graph modeling, and CACL—this framework provides an effective solution for few-shot hyperspectral classification.

The input HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B denote the height, width, and number of spectral bands, respectively, first undergoes preprocessing via a lightweight CNN denoise module, the structure of which is shown in Fig. 2. This module consists of two sequential processing blocks, each containing batch normalization, 1×1 convolution, and LeakyReLU activation. It projects the original spectral features into a unified D-dimensional feature space, obtaining $\mathbf{X}_{\text{denoise}} \in \mathbb{R}^{H\times W\times D}$ as standardized input feature representations for the subsequent CNN and GCN branches.

B. SFEM

Standard CNNs operating in the raw spectral domain are limited by their local receptive fields, making it challenging to capture global spectral dependencies, especially when training samples are scarce. In contrast, frequency-domain transformation decomposes signals into global basis functions, offering a holistic view of the signal. Research indicates that analyzing features in the frequency domain can effectively isolate robust, domain-invariant components, thereby significantly improving the model's generalization capability [53]. Therefore, in few-shot scenarios, introducing frequency-domain features not only extracts more stable spectral fingerprints but also serves as an effective regularization mechanism, enhancing the model's discriminative power. Based on this, we propose the SFEM for frequency-domain feature extraction.

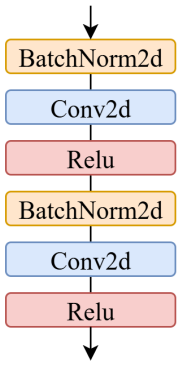


Fig. 2. Structure of the CNN denoise.

The overall architecture of the proposed SFEM is illustrated in Fig. 3. The preprocessed feature $\mathbf{X}_{\text{denoise}}$ is first flattened along the spatial dimensions to form a 2-D matrix $\mathbf{X}_{\text{flat}} \in \mathbb{R}^{N\times D}$, where $N = H \times W$ represents the total number of pixels. Subsequently, the flattened feature is transformed from the spatial domain to the frequency domain using the

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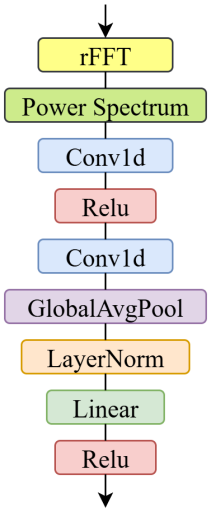


Fig. 3. Structure of the proposed SFEM.

real-valued fast Fourier transform (rFFT), yielding a complex frequency spectrum

$$\mathbf{X}_f = \mathrm{rFFT}\left(\mathbf{X}_{\mathrm{flat}}\right). \tag{1}$$

Subsequently, the power spectral density (PSD) is computed to characterize the signal's energy distribution

$$\mathbf{P} = \left|\mathbf{X}_f\right|^2. \tag{2}$$

Logarithmic compression is applied to the power spectrum to enhance discrimination in lower frequencies

$$\mathbf{P}_{\log} = \log\left(1 + \mathbf{P}\right). \tag{3}$$

The compressed power spectrum is then fed into a 1-D convolutional network (1D-CNN) for feature learning to extract discriminative frequency-domain features, $\mathbf{F}_{\mathrm{freq}}$

$$\mathbf{F}_{\mathrm{freq}} = \text{1D-CNN}\left(\mathbf{P}_{\log}\right) \in \mathbb{R}^{N\times d} \tag{4}$$

where d is the dimensionality of the frequency-domain features used for subsequent feature fusion.

The CNN branch employs a CNN-spatial module to extract local spatial features. As illustrated in Fig. 4, the module first applies a pointwise convolution (1×1) for cross-channel information fusion and dimensionality adjustment, followed by a depthwise convolution that performs independent spatial filtering on each channel. This decomposition approach significantly reduces computational complexity while maintaining representational capacity. Each convolutional layer is equipped with batch normalization and LeakyReLU activation functions to ensure stable training and nonlinear transformation. The CNN-spatial module processes $\mathbf{X}_{\mathrm{denoise}}$ through multiple such processing units to extract local spatial features, ultimately producing $\mathbf{F}_{\mathrm{spa}} \in \mathbb{R}^{N\times d}$.

Finally, the spectral frequency-domain features $\mathbf{F}_{\mathrm{freq}}$ are fused with the local spatial features $\mathbf{F}_{\mathrm{spa}}$ using a fixed weighting scheme. The fusion employs a weight parameter α to balance the contributions of both feature types (the impact of different α values on classification performance is systematically analyzed in the ablation study section), forming a

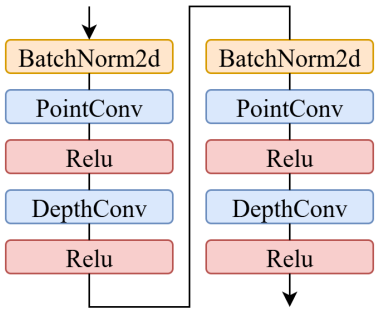


Fig. 4. Structure of the spatial CNN.

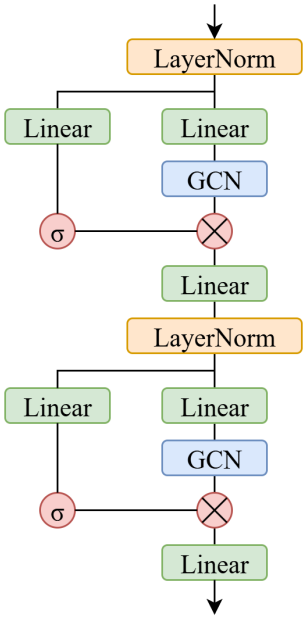


Fig. 5. Structure of the gated GCN.

combined representation $\mathbf{F}_{\mathrm{cnn}} \in \mathbb{R}^{N\times d}$ that integrates global spectral characteristics with local spatial details

$$\mathbf{F}_{\mathrm{cnn}} = \alpha\mathbf{F}_{\mathrm{spa}} + (1-\alpha)\,\mathbf{F}_{\mathrm{freq}}. \tag{5}$$

C. AMGCM

To construct the graph nodes efficiently, the LDA-SLIC [41] algorithm is adopted. This method specifically addresses the high dimensionality of HSI by incorporating linear discriminant analysis (LDA) to project spectral features into a lower-dimensional discriminant subspace. Subsequently, pixels are clustered into compact superpixels based on a distance metric that combines the spectral distance in the projected subspace with the spatial Euclidean distance. This ensures that the generated superpixels align well with object boundaries while reducing computational costs. In this work, this algorithm is employed to generate multigranularity superpixel segmentation results, constructing hierarchical graph structures. For each scale level $s \in \{1, 2, \ldots, s_{\max}\}$, where $s_{\max}$ is the maximum scale, the image is segmented into $S_s = N/s$ superpixel regions, with each superpixel serving as a node in the graph.

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To comprehensively characterize the relationships between superpixels, we construct two complementary types of adjacency matrices for each scale.

1) Euclidean distance adjacency matrix $\mathbf{A}_s^e$, whose elements are derived from the Euclidean distance between superpixel node features transformed via a Gaussian kernel

$$\left[\mathbf{A}_s^e\right]_{ij} = \exp\left(-\gamma \left\|\mathbf{f}_i - \mathbf{f}_j\right\|_2^2\right). \tag{6}$$

Here, $\mathbf{f}_i$ and $\mathbf{f}_j$ are the average spectral feature vectors of all pixels within superpixels i and j, respectively, and γ is a scale parameter controlling the weight decay rate.

2) Binary adjacency matrix $\mathbf{A}_s^b$, which simply encodes the spatial connectivity relationship between superpixels

$$\left[\mathbf{A}_s^b\right]_{ij} = \begin{cases} 1, & \text{if node } i \text{ is adjacent to node } j \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

To ensure that each node retains its own feature information during feature aggregation and to improve numerical stability, we perform symmetric normalization with self-loops on both adjacency matrices

$$\hat{\mathbf{A}} = \mathbf{A} + \lambda \mathbf{I} \tag{8}$$

$$\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-1/2} \tag{9}$$

where $\mathbf{A}$ represents the adjacency matrix $\mathbf{A_s^e}$ or $\mathbf{A_s^b}$, λ is the self-loop weight, $\hat{\mathbf{A}}$ is the matrix after adding self-loops, $\hat{\mathbf{D}}$ is the degree matrix of $\hat{\mathbf{A}}$, and $\tilde{\mathbf{A}}$ is the normalized result matrix. The normalized matrices of the adjacency matrices $\mathbf{A_s^e}$ and $\mathbf{A_s^b}$ are denoted as $\tilde{\mathbf{A}}_s^e$ and $\tilde{\mathbf{A}}_s^b$, respectively, and are used for the subsequent graph convolution computations.

During the feature processing stage, the input image is processed through the CNN denoise module to obtain $\mathbf{X}_{\text{denoise}}$, which is then flattened into $\mathbf{X}_{\text{flat}}$. Using the pixel-superpixel association matrix $\mathbf{Q}_s \in \mathbb{R}^{N \times S_s}$ (which indicates which superpixel each pixel belongs to) at scale s, we aggregate the pixel features into the superpixel space:

$$\mathbf{H}_s^{\text{in}} = \mathbf{Q}_s^\top \mathbf{X}_{\text{flat}} \in \mathbb{R}^{S_s \times D}. \tag{10}$$

Subsequently, a two-level Gated GCN is employed to enhance feature representation, as illustrated in Fig. 5. First, layer normalization is applied to the node features $\mathbf{H}_s^{\text{in}}$, which are then linearly transformed and split into three components (parameterized by $\mathbf{W}_1^e$): the gating component $\mathbf{G}_s^1$, the identity component $\mathbf{I}_s^1$, and the graph convolution component $\mathbf{C}_s^1$. A selective graph convolution mechanism is adopted, where only the $\mathbf{C}_s^1$ component undergoes feature propagation via the graph convolution operation, while the $\mathbf{I}_s^1$ and $\mathbf{G}_s^1$ components are reserved for residual connection and gated feature fusion, respectively. This approach effectively reduces computational complexity while maintaining the model's expressive power

$$\mathbf{C}_s' = \text{ReLU}\left(\tilde{\mathbf{A}}_s^e \mathbf{C}_s^1 \mathbf{W}_g^e\right) \tag{11}$$

where $\tilde{\mathbf{A}}_s^e$ is the normalized Euclidean distance adjacency matrix and $\mathbf{W}_g^e$ is the weight parameter for graph convolution. Feature fusion is then achieved through a gating mechanism that combines the graph convolution results with the gating and identity components

$$\mathbf{H}_s^1 = \mathbf{H}_s^{\text{in}} + \text{DropPath}\left(\mathbf{W}_2^e \left(\text{GELU}\left(\mathbf{G}_s^1\right) \odot \left[\mathbf{I}_s^1, \mathbf{C}_s'\right]\right)\right). \tag{12}$$

Here, $\mathbf{W}_2^e$ is the learnable weight matrix for gated fusion, $\odot$ denotes element-wise multiplication, and $[\cdot,\cdot]$ represents the concatenation operation. The gating mechanism, regulated by the GELU activation function, modulates the information flow and enhances the model's nonlinear expressive capability.

The second level takes the output $\mathbf{H}_s^1$ from the first level as its input. It repeats the projection and splitting process (parameterized by $\mathbf{W}_1^b$) to obtain new gating component $\mathbf{G}_s^2$, identity component $\mathbf{I}_s^2$, and convolution component $\mathbf{C}_s^2$. The graph convolution is then applied using the binary adjacency matrix $\tilde{\mathbf{A}}_s^b$ to capture spatial connectivity

$$\mathbf{C}_s'' = \text{ReLU}\left(\tilde{\mathbf{A}}_s^b \mathbf{C}_s^2 \mathbf{W}_g^b\right). \tag{13}$$

Finally, the output of the second level is obtained through the gated fusion with a residual connection to $\mathbf{H}_s^1$

$$\mathbf{H}_s^2 = \mathbf{H}_s^1 + \text{DropPath}\left(\mathbf{W}_2^b \left(\text{GELU}\left(\mathbf{G}_s^2\right) \odot \left[\mathbf{I}_s^2, \mathbf{C}_s''\right]\right)\right) \tag{14}$$

where $\mathbf{W}_g^b$ and $\mathbf{W}_2^b$ are the learnable weight matrices for the second-level graph convolution and gated fusion, respectively.

The enhanced superpixel features are then mapped back to the pixel space, transforming node-level features to pixel-level features

$$\mathbf{H}_s^{\text{out}} = \mathbf{Q}_s \mathbf{H}_s^2. \tag{15}$$

Through linear transformation (parameterized by a learnable weight matrix $\mathbf{W}_3$) and batch normalization, the feature dimension is reduced to d, and the LeakyReLU activation function is applied to introduce nonlinearity

$$\mathbf{F}_s = \text{LeakyReLU}\left(\text{BatchNorm}\left(\mathbf{H}_s^{\text{out}} \mathbf{W}_3\right)\right). \tag{16}$$

Based on the learnable parameters β_s, the fusion weight w_s for each scale s is calculated using the softmax function to achieve adaptive multiscale feature fusion

$$w_s = \frac{\exp\left(\beta_s\right)}{\sum_{i=1}^{s_{\max}} \exp\left(\beta_i\right)}. \tag{17}$$

Finally, the features from all scales are fused through weighted summation to obtain the final output $\mathbf{F}_{\text{gcn}} \in \mathbb{R}^{N \times d}$ of the graph convolution branch

$$\mathbf{F}_{\text{gcn}} = \sum_{s=1}^{s_{\max}} w_s \cdot \mathbf{F}_s. \tag{18}$$

D. CACL

To achieve feature representation alignment between the CNN and GCN branches, this article designs a CACL mechanism. This mechanism constructs positive sample pairs by pairing the features extracted from the same pixel by the CNN and GCN branches, encouraging the dual branches to learn consistent feature representations. However, establishing such cross-architecture alignment poses a significant challenge due to the inherent heterogeneity of the extracted features. The CNN branch processes data in Euclidean space, focusing on local spatial details, whereas the GCN branch operates in non-Euclidean space, capturing topological structures. This fundamental difference results in a heterogeneous distribution gap between the two feature spaces. To bridge this gap, the proposed mechanism leverages contrastive learning as a core

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alignment strategy, utilizing specific projection heads to map these heterogeneous features into a shared latent space.

Specifically, for each pixel in the image, the feature $\mathbf{F}_{\text{cnn}}$ extracted by the CNN branch and the feature $\mathbf{F}_{\text{gcn}}$ extracted by the GCN branch form a positive sample pair. Features from different pixels within the same training batch naturally constitute negative sample pairs.

First, high-dimensional features are mapped to a low-dimensional contrastive space through two independent projection heads

$$\mathbf{Z}_c = \text{Proj}_c\left(\mathbf{F}_{\text{cnn}}\right) \in \mathbb{R}^{N\times k} \tag{19}$$

$$\mathbf{Z}_g = \text{Proj}_g\left(\mathbf{F}_{\text{gcn}}\right) \in \mathbb{R}^{N\times k} \tag{20}$$

where Proj_c and Proj_g are the projection heads for the CNN and GCN branches, respectively, implemented by multilayer perceptrons that map high-dimensional features to a k-dimensional contrastive space. The projected features undergo L2 normalization to ensure similarity computation is performed on a unit hypersphere, which is crucial for the stability of contrastive learning.

The normalized temperature-scaled cross-entropy (NT-Xent) loss function is employed for contrastive learning. For a training batch containing M pixels, the contrastive loss for the ith pixel is calculated as

$$\mathcal{L}_{\text{cont}}^{(i)} = -\log\left[\frac{N^{(i)}}{D^{(i)}}\right] \tag{21}$$

$$N^{(i)} = \exp\left(\text{sim}\left(\mathbf{Z}_c^{(i)}, \mathbf{Z}_g^{(i)}\right)/\tau\right) \tag{22}$$

$$D^{(i)} = \sum_{j=1}^{M} \exp\left(\text{sim}\left(\mathbf{Z}_c^{(i)}, \mathbf{Z}_g^{(j)}\right)/\tau\right) + \sum_{\substack{j=1\\ j\neq i}}^{M} \exp\left(\text{sim}\left(\mathbf{Z}_c^{(i)}\mathbf{Z}_c^{(j)}\right)/\tau\right) \tag{23}$$

where $N^{(i)}$ represents the positive pair term, which measures the similarity between the CNN and GCN features of the same pixel i, the temperature parameter τ controls the sharpness of the similarity distribution, with smaller values emphasizing harder negative samples. $D^{(i)}$ is the denominator term that includes two components: the first sum computes the similarities between the CNN feature of pixel i and the GCN features of all pixels in the batch, while the second sum computes the similarities between the CNN feature of pixel i and the CNN features of all other pixels in the batch. This formulation ensures that the model learns to distinguish between positive pairs (same pixel and different branches) and negative pairs (different pixels, same or different branches). The final contrastive loss is averaged over all pixels in the batch

$$\mathcal{L}_{\text{cont}} = \frac{1}{M}\sum_{i=1}^{M}\mathcal{L}_{\text{cont}}^{(i)}. \tag{24}$$

This article adopts a dual-stage training strategy combining pre-training and fine-tuning, which aims to enable the model to first learn general feature representations with strong generalization capabilities, and then optimize for specific classification tasks.

In the pre-training stage, the model is trained in an unsupervised manner, with the objective of encouraging the CNN and GCN branches to learn consistent feature representations without utilizing any sample labels. The training process is optimized solely using the contrastive loss function $\mathcal{L}_{\text{cont}}$ based on CACL. This loss function learns category-agnostic, discriminative foundational features by maximizing the consistency of feature representations for the same pixel across the two branches, while simultaneously pushing apart the feature representations of different pixels.

After pre-training is completed, the model enters the supervised fine-tuning stage. First, the features extracted by the CNN and GCN branches are weighted and fused to obtain the fused feature $\mathbf{F}_{\text{fusion}}$, with δ as the weight parameter

$$\mathbf{F}_{\text{fusion}} = \delta\cdot\mathbf{F}_{\text{cnn}} + (1-\delta)\cdot\mathbf{F}_{\text{gcn}}. \tag{25}$$

Subsequently, the fused features are passed through a linear classifier for category prediction. Training is conducted using data with true labels to optimize the final classification performance. For a batch containing K pixel samples, the cross-entropy loss function is employed

$$\mathcal{L}_{\text{sup}} = -\frac{1}{K}\sum_{i=1}^{K}\sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{26}$$

where C is the total number of classes in the dataset, $y_{i,c}$ is the smoothed ground-truth label of the ith pixel sample corresponding to class c, and $p_{i,c}$ is the probability predicted by the model that the ith pixel sample belongs to class c. The entire fine-tuning process optimizes the model parameters by minimizing this loss function.

III. Experiment

A. Data Description

1) *WHU-Hi-HanChuan (WHU) Dataset:* This dataset was collected in 2017 using the Headwall Nano-Hyperspec imaging sensor in HanChuan City, Hubei Province, China. It contains 1217 × 303 pixels with a spatial resolution of 0.109 m and a wavelength range from 400 to 1000 nm, comprising 274 spectral bands. The dataset includes 257 530 labeled pixels, divided into 16 land cover classes, as detailed in Table I. The pseudo-color and ground-truth images can be referenced in Fig. 6.
2) *Indian Pines (IP) Dataset:* This dataset was collected in 1992 by the airborne visible/infrared imaging spectrometer (AVIRIS) over the northwestern region of Indiana, USA. It contains 145 × 145 pixels with a spatial resolution of 20 m and a wavelength range from 400 to 2500 nm, including 200 spectral bands. The dataset has 10 249 labeled pixels, categorized into 16 land cover classes, as shown in Table I. The pseudo-color and ground-truth images are available in Fig. 7.
3) *University of Pavia (UP) Dataset:* This dataset was collected in 2002 using the reflective optics imaging spectrometer (ROIS) sensor in the Pavia region of Italy. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m and a wavelength range from 400 to 2500 nm, containing 103 spectral bands. The dataset includes

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TABLE I
DETAILS OF THE THREE EXPERIMENTAL DATASETS

WHU-Hi-HanChuan				Indian Pines				University of Pavia			
ID	**Category**	**Train**	**Test**	**ID**	**Category**	**Train**	**Test**	**ID**	**Category**	**Train**	**Test**
1	Strawberry	30	44705	1	Alfalfa	30	16	1	Asphalt	30	6601
2	Cowpea	30	22723	2	Corn-notill	30	1398	2	Meadows	30	18619
3	Soybean	30	10257	3	Corn-mintill	30	800	3	Gravel	30	2069
4	Sorghum	30	5323	4	Corn	30	207	4	Trees	30	3034
5	Water spinach	30	1170	5	Grass-pasture	30	453	5	Painted metal sheets	30	1315
6	Watermelon	30	4503	6	Grass-trees	30	700	6	Bare soil	30	4999
7	Greens	30	5873	7	Grass-pasture-mowed	15	13	7	Bitumen	30	1300
8	Trees	30	17948	8	Hay-windrowed	30	448	8	Self-blocking bricks	30	3652
9	Grass	30	9439	9	Oats	15	5	9	Shadows	30	917
10	Red roof	30	10486	10	Soybean-notill	30	942				
11	Gray roof	30	16881	11	Soybean-mintill	30	2425				
12	Plastic	30	3649	12	Soybean-clean	30	563				
13	Bare soil	30	9086	13	Wheat	30	175				
14	Road	30	18530	14	Woods	30	1235				
15	Bright object	30	1106	15	Building-grass-trees-drives	30	356				
16	Water	30	75371	16	Stone-steel-towers	30	63				
Total		480	257050			450	9799			270	42506

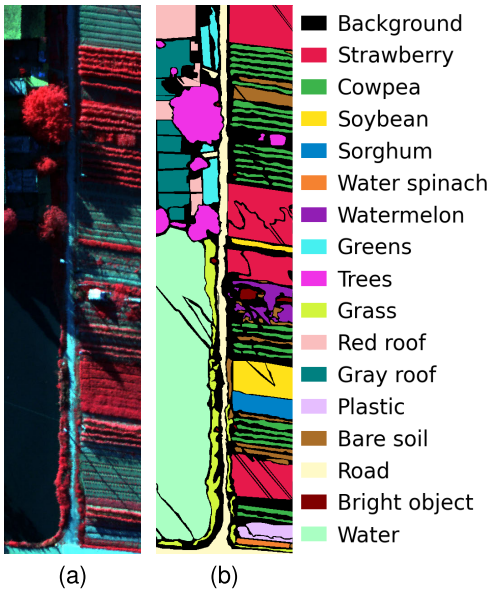


Fig. 6. WHU dataset. (a) False color composite (bands R: 200, G: 120, B: 50). (b) Ground-truth map.

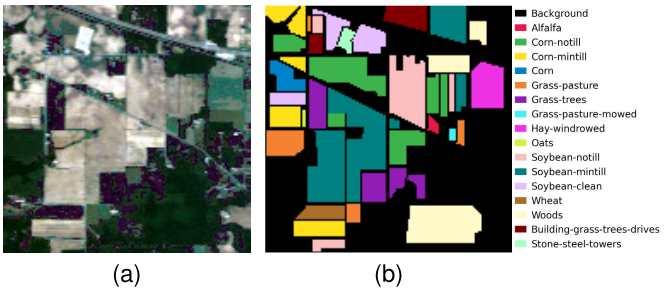


Fig. 7. IP dataset. (a) False color composite (bands R: 28, G: 19, B: 10). (b) Ground-truth map.

42 776 labeled pixels, divided into nine land cover classes, as detailed in Table I. The pseudo-color and ground-truth images can be referenced in Fig. 8.

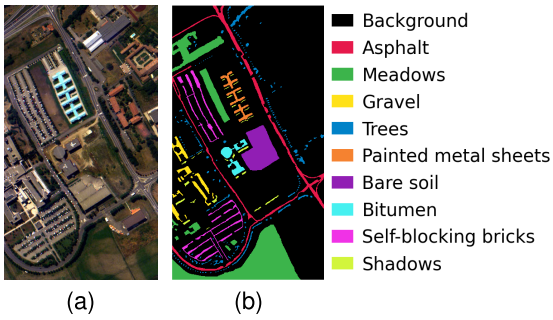


Fig. 8. UP dataset. (a) False color composite (bands R: 55, G: 34, B: 3). (b) Ground-truth map.

B. Experimental Settings

We conducted the experiments using the PyTorch deep learning framework on a hardware platform equipped with an Intel Xeon Gold 5317 CPU and an NVIDIA GeForce RTX 3090 GPU. The optimization was performed using the Adam optimizer with an initial learning rate of 10^{-3} and a weight decay of 10^{-3} for regularization. The training process consisted of 100 epochs for pre-training and 300 epochs for fine-tuning. Regarding the graph construction, the superpixel scales were set to 700, 1000, and 1400, respectively. For the adjacency calculation, the scale parameter γ was set to 10^{-2} to control the weight decay rate of the Gaussian kernel, and the self-loop weight λ was fixed at 15 to reinforce node self-importance. Additionally, the weight α for fusing spatial and frequency features was set to 0.85, while the branch balancing weight δ was set to 0.4. Three public HSI datasets were selected: WHU, IP, and UP. To evaluate the model's performance under few-shot conditions, a restricted training sample strategy was adopted: 30 labeled samples were randomly selected per class; if a class had fewer than 30 samples, 15 were chosen, with the remaining samples used as the test set. The specific numbers of training and test samples are detailed in Table I. Model performance was comprehensively

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TABLE II
CLASSIFICATION RESULTS (%) OF THE WHU DATASET. BEST RESULTS ARE IN BOLDFACE

Class	HybridSN	RSSAN	A2S2K	HiBiCo	AMGCFN	GiGCN	NESSGGCN	CACL
1	94.97±0.01	79.42±6.65	95.39±0.30	87.55±3.25	95.70±1.54	**97.13±1.43**	95.37±1.62	95.63±1.83
2	90.27±0.02	55.37±5.43	**95.73±3.06**	44.18±3.76	90.13±2.53	87.55±1.70	87.20±2.74	90.64±2.36
3	92.94±0.01	47.92±13.96	79.35±6.96	55.21±7.88	98.08±0.79	97.68±1.41	97.16±1.04	**99.55±0.18**
4	81.55±0.03	80.16±12.39	92.44±3.62	76.78±6.27	**98.10±1.05**	96.54±1.91	97.75±1.28	97.68±0.81
5	35.62±0.02	95.15±1.59	57.27±16.92	92.83±6.41	99.35±1.21	**99.99±0.03**	98.70±1.25	99.32±0.44
6	66.50±0.03	25.59±7.96	55.67±16.20	25.77±6.21	95.50±3.16	**96.81±2.72**	92.41±2.92	95.65±2.64
7	92.27±0.02	84.14±6.12	64.21±7.47	79.10±5.49	97.35±0.87	**99.47±0.40**	96.54±3.03	93.86±3.76
8	**97.32±0.01**	36.83±9.92	92.85±2.06	50.75±6.12	88.72±2.29	83.04±4.19	86.15±4.39	90.76±4.65
9	73.07±0.02	36.40±2.81	71.48±7.89	43.26±6.74	**90.73±3.45**	87.15±4.87	81.88±3.27	87.97±3.84
10	83.50±0.03	70.07±8.81	95.63±2.63	92.80±1.58	97.03±0.95	96.23±2.36	**98.57±0.45**	97.22±1.35
11	94.76±0.01	62.94±7.79	72.16±6.16	81.28±7.06	95.27±3.08	96.74±1.15	93.60±4.54	**96.73±1.99**
12	63.08±0.03	62.12±6.95	72.20±13.24	76.72±9.62	99.16±1.15	98.59±2.21	97.18±3.21	**99.35±0.57**
13	72.24±0.04	55.40±3.75	53.89±8.57	51.81±5.35	80.86±10.40	76.64±4.70	80.67±5.34	**85.31±4.20**
14	69.72±0.02	78.90±3.38	78.77±10.49	58.42±6.38	86.60±6.80	86.48±3.93	91.56±1.71	**92.01±2.91**
15	70.07±0.05	82.11±6.33	70.71±6.34	82.48±5.96	**94.20±2.29**	83.28±4.33	89.82±5.44	91.97±3.36
16	99.31±0.00	94.69±1.34	**99.90±0.07**	92.09±2.42	97.62±0.91	97.59±1.19	97.33±1.00	99.02±0.42
OA	88.78±0.01	72.63±2.75	85.86±1.36	74.35±0.76	94.18±0.63	93.54±0.48	93.41±0.84	**95.29±0.46**
AA	87.43±0.02	65.45±3.24	77.98±1.85	58.38±0.22	94.03±0.54	92.56±0.58	92.62±0.83	**94.54±0.45**
Kappa	86.92±0.01	68.26±3.16	83.62±1.55	70.42±0.84	93.22±0.73	92.47±0.56	92.31±0.97	**94.49±0.54**

evaluated using three widely adopted metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. OA represents the proportion of correctly classified test samples out of the total test samples; AA is the arithmetic mean of the classification accuracies of all classes; and the Kappa coefficient measures the agreement between the classification results and the ground truth, accounting for the effect of random classification. Higher values for all metrics indicate better classification performance.

For comparative methods, to validate the effectiveness of the proposed approach, seven state-of-the-art HSI classification methods were selected for comparison, covering three categories: CNNs, contrastive learning, and GCNs. Specifically, CNN-based methods include HybridSN [54], A2S2K-ResNet [55], and RSSAN [56]; the contrastive learning method is HiBiCo [57]; and GCN-based methods include AMGCFN [58], GiGCN [59], and NESSGGCN [41]. All comparative experiments employed the restricted training sample strategy. To mitigate randomness due to the random splitting of training samples and model parameter initialization, each experiment was independently repeated five times, and the final reported results are the average values and standard deviations of the metrics obtained from these five runs.

C. Classification Results

To evaluate the effectiveness of the proposed method, a series of experiments was conducted under few-shot conditions on the WHU, IP, and UP datasets.

1) Results on WHU: Table II presents the quantitative classification results on the WHU dataset with limited training samples. Our proposed CACL method achieved the best performance, with OA, AA, and Kappa coefficient reaching 95.29%, 94.54%, and 94.49%, respectively. A comparative analysis further highlights the effectiveness of our method. First, CNN-based methods performed poorly under few-shot conditions (none achieved an OA above 90%). It is worth noting that while these methods perform well with sufficient training data, their limitations are magnified when training samples are extremely scarce: the local receptive field struggles to capture the long-range contextual dependencies necessary for distinguishing similar crops in agricultural scenes, leading to significant performance degradation. In contrast, our method effectively captures contextual information between pixels through the graph structure modeling of GCN, addressing this issue. Notably, although the contrastive learning method HiBiCo is effective on other datasets, its performance on this dataset (OA: 74.35%) indicates that contrastive learning lacking structural priors tends to suffer from reduced generalization under very few-shot conditions. Our method ensures more stable feature alignment for contrastive learning through the robust graph structural prior provided by GCN. Furthermore, although GCN-based methods (e.g., AMGCFN and GiGCN) can capture global relationships between pixels through graph structures, their lack of explicit constraints on feature discriminativity leads to suboptimal performance. In comparison, our CACL mechanism significantly enhances the model's ability to recognize subtle spectral differences by introducing frequency-domain enhancement. The frequency-domain enhancement captures the global morphological characteristics of spectral signals, while the contrastive learning mechanism ensures feature consistency and discriminativity. The synergistic advantages brought by GCN's structural modeling capability and the discriminative constraints of contrastive learning are reflected in the detailed results for each category. The proposed method achieved the highest accuracy in 5 out of the 16 classes and demonstrated excellent stability. Compared to the high variance shown by other baseline methods, our method proves to be more robust

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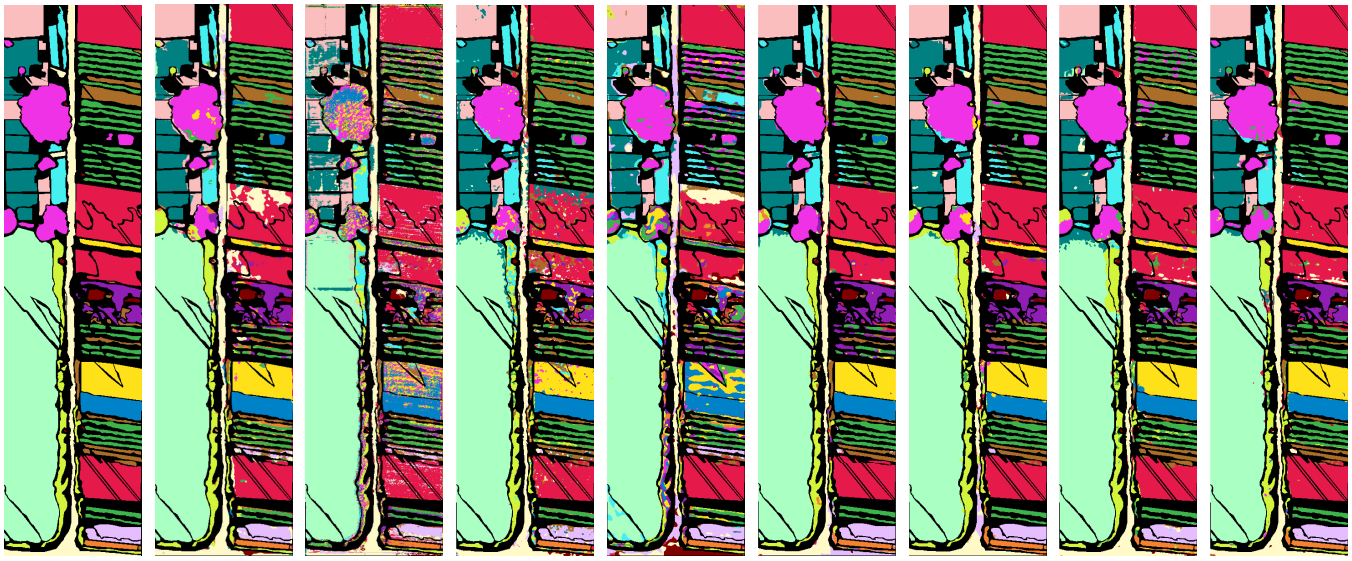


Fig. 9. Classification maps of different methods on the WHU dataset. (a) Ground truth. (b) HybridSN. (c) RSSAN. (d) A2S2K. (e) HiBiCo. (f) AMGCFN. (g) GiGCN. (h) NESSGGCN. (i) CACL.

in few-shot scenarios and less sensitive to random splits of training samples.

The qualitative classification results intuitively corroborate the conclusions of quantitative analysis. As shown in Fig. 9, the classification maps generated by the proposed CACL method significantly outperform existing baseline methods in both spatial coherence and category discriminability. CNN-based methods (particularly RSSAN) exhibit severe "salt-and-pepper noise" in their results, with extensive fragmented misclassifications occurring at boundaries between spectrally similar vegetation categories. This highlights the limitations of their local modeling capabilities under extremely limited sample conditions. The purely contrastive learning method HiBiCo, lacking spatial structure priors, suffers from diffuse errors in atypical regions and insufficient generalization ability. Although GCN-based methods achieve smoother classification results through graph structures, they still exhibit boundary blurring when distinguishing highly similar crops. In contrast, the proposed method effectively captures subtle spectral differences through frequency-domain enhancement and CACL mechanisms. This results in clearer boundaries and purer homogeneous regions across various land cover types, intuitively demonstrating the exceptional synergistic effect between the robust structural prior provided by GCN and the strong discriminative power conferred by contrastive learning.

2) Results on IP: Table III presents the quantitative classification results on the IP dataset with limited training samples. Our proposed method achieved the best performance, with OA, AA, and Kappa coefficient reaching 96.73%, 98.25%, and 96.25%, respectively. CNN-based methods showed significant variation in performance under few-shot conditions: RSSAN (OA: 70.07%) performed poorly, as its complex structure is difficult to train effectively with limited samples; whereas HybridSN (OA: 88.69%) and A2S2K (OA: 90.34%) reached competitive baseline levels. Notably, the contrastive learning method HiBiCo (OA: 90.15%) also performed well, but its performance still lagged behind the leading GCN methods. Among GCN methods, AMGCFN (OA: 94.00%) and NESSGGCN (OA: 95.21%) formed the top performance tier, confirming the advantage of graph structure modeling for this task. Our method further increased the OA to 96.73%, achieving a significant improvement over existing strong baselines. This result is consistent with the conclusions from the WHU dataset, further validating the ability of our CACL framework to stably surpass the performance bottlenecks of traditional GCNs across different scenarios. Additionally, our method demonstrated exceptional stability, with the second smallest standard deviations for OA, AA, and Kappa among all methods, and achieved a very high AA of 98.25%, highlighting its strong capability in handling class imbalance. More convincingly, the method achieved 100% classification accuracy for five land-cover classes (classes 1, 7, 8, 9, 16), fully demonstrating its generalization ability and reliability across different scenarios.

The qualitative classification results on the IP dataset align with the conclusions drawn from the WHU dataset and further highlight the generalization capability of the proposed method. As shown in Fig. 10, compared to the already high-performing GCN baseline methods, the classification maps generated by our approach exhibit sharper boundaries for ground objects and more accurate identification of small-scale, sparsely distributed features (such as rare classes like oats and hay-windrowed). This effectively mitigates the slight blurring and class adhesion issues observed in existing GCN methods within detailed regions. Most notably, the proposed method achieves near-perfect classification for several of the most challenging rare categories, producing classification regions with high internal purity and no fragmentation. This observation aligns closely with the quantitative results, where five categories achieved 100% accuracy and an AA of 98.25%, providing intuitive evidence of the strong discriminative power

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TABLE III
CLASSIFICATION RESULTS (%) OF THE IP DATASET. BEST RESULTS ARE IN BOLDFACE

Class	HybridSN	RSSAN	A2S2K	HiBiCo	AMGCFN	GiGCN	NESSGGCN	CACL
1	88.89±0.00	100.00±0.00	81.64±6.70	100.00±0.00	100.00±0.00	99.13±1.06	96.25±5.00	**100.00±0.00**
2	84.14±0.19	45.02±5.34	91.61±3.92	85.69±5.52	89.44±4.99	79.88±5.40	89.04±5.38	**93.61±3.23**
3	77.59±0.21	63.03±2.18	85.76±3.25	81.74±8.71	95.96±2.44	90.64±3.91	**97.45±1.51**	96.02±1.67
4	99.04±0.00	75.47±6.83	84.45±6.53	99.52±0.68	**99.76±0.50**	89.41±9.91	98.94±0.94	99.71±0.58
5	98.71±0.10	73.18±6.62	**99.04±1.35**	94.11±1.87	95.08±3.37	74.53±5.12	96.51±4.04	96.07±1.00
6	98.07±0.07	77.94±3.98	94.58±2.47	99.51±0.66	98.34±1.06	95.27±3.46	**99.91±0.11**	99.80±0.07
7	92.86±0.00	100.00±0.00	56.94±7.08	100.00±0.00	100.00±0.00	93.21±3.37	100.00±0.00	**100.00±0.00**
8	96.08±0.00	97.60±1.50	99.02±1.22	99.62±1.14	99.93±0.20	99.83±0.13	100.00±0.00	**100.00±0.00**
9	11.27±0.29	100.00±0.00	51.47±34.36	100.00±0.00	100.00±0.00	22.00±27.50	100.00±0.00	**100.00±0.00**
10	82.00±0.25	63.47±9.26	75.46±1.37	85.74±6.75	91.65±3.89	75.72±4.34	92.14±5.16	**94.52±2.21**
11	91.85±0.14	69.61±5.47	**96.61±0.94**	88.55±3.78	90.78±2.99	89.92±1.48	92.81±1.18	95.74±1.75
12	**99.16±0.00**	61.18±4.52	81.40±8.69	83.98±4.25	92.24±4.02	92.12±3.16	97.66±1.23	97.98±1.08
13	86.38±0.38	100.00±0.00	89.04±4.56	**100.00±0.00**	99.66±0.73	99.90±0.20	99.66±0.28	99.77±0.28
14	97.13±0.07	86.41±1.82	98.89±0.84	96.48±2.26	98.12±1.65	98.85±2.81	98.82±2.00	**99.00±1.30**
15	69.75±0.18	82.51±7.69	83.80±2.35	88.82±5.58	99.24±0.87	98.32±0.17	**100.00±0.00**	99.83±0.34
16	83.34±0.61	99.27±0.89	80.84±5.16	98.25±2.40	98.89±1.43	97.85±0.00	99.68±0.63	**100.00±0.00**
OA	88.69±0.11	70.07±0.96	90.34±0.84	90.15±1.74	94.00±0.76	89.08±1.00	95.21±1.50	**96.73±0.50**
AA	94.77±0.08	80.92±0.79	84.41±2.10	83.34±0.19	96.82±0.29	87.29±1.64	97.43±1.18	**98.25±0.29**
Kappa	87.14±0.13	65.95±1.09	89.03±0.94	88.77±1.96	93.15±0.86	87.60±1.14	94.52±1.70	**96.25±0.57**



Fig. 10. Classification maps of different methods on the IP dataset. (a) Ground truth. (b) HybridSN. (c) RSSAN. (d) A2S2K. (e) HiBiCo. (f) AMGCFN. (g) GiGCN. (h) NESSGGCN. (i) CACL.

TABLE IV
CLASSIFICATION RESULTS (%) OF THE UP DATASET. BEST RESULTS ARE IN BOLDFACE

Class	HybridSN	RSSAN	A2S2K	HiBiCo	AMGCFN	GiGCN	NESSGGCN	CACL
1	88.68±0.05	67.28±3.98	96.41±3.10	93.51±4.71	97.55±1.76	98.26±1.13	96.35±2.01	**98.79±1.02**
2	98.61±0.01	88.36±2.72	98.14±0.69	92.65±4.99	97.38±1.91	99.18±0.49	94.35±4.17	**99.25±0.58**
3	88.62±0.08	55.34±4.66	74.49±7.03	97.19±2.17	99.10±1.04	**99.93±0.09**	99.42±0.57	99.15±0.66
4	94.81±0.03	83.46±2.18	**98.66±0.86**	94.13±2.18	97.13±1.06	87.09±2.19	96.62±1.33	97.03±1.30
5	92.79±0.08	98.56±0.87	99.92±0.06	98.71±1.68	99.93±0.11	99.99±0.02	100.00±0.00	**100.00±0.00**
6	96.84±0.01	58.09±4.49	93.59±3.33	98.77±1.71	**100.00±0.00**	99.76±0.47	96.54±2.05	99.77±0.24
7	98.51±0.04	68.20±6.76	83.13±4.29	99.76±0.30	99.80±0.30	**99.93±0.05**	99.49±0.37	99.92±0.12
8	90.27±0.09	61.55±8.62	82.12±5.46	96.85±3.20	98.82±0.45	98.67±0.79	97.74±0.50	**99.72±0.22**
9	76.82±0.15	96.36±1.89	98.17±1.34	97.06±1.68	97.09±0.52	85.90±3.21	99.93±0.05	**99.96±0.05**
OA	94.67±0.03	77.14±0.85	93.95±0.79	94.69±2.36	98.05±0.92	97.99±0.30	96.07±1.62	**99.17±0.24**
AA	93.08±0.05	75.24±1.50	91.62±0.32	94.70±0.04	98.53±0.43	96.52±0.41	97.83±0.55	**99.29±0.16**
Kappa	92.91±0.03	69.79±1.13	91.99±1.04	93.08±3.00	97.43±1.20	97.33±0.39	94.85±2.08	**98.91±0.31**

and robustness against extreme class imbalance endowed by the CACL mechanism.

3) Results on UP: Table IV presents the quantitative classification results on the UP dataset with limited training samples. Our proposed method achieved the best performance, with OA, AA, and Kappa coefficient reaching 99.17%, 99.29%, and 98.91%, respectively. Its performance is near-perfect and significantly leads all baseline models. On this dataset, the performance gap between different method categories is clearer. Among CNN-based methods, RSSAN (OA: 77.14%) again performed poorly due to its high model complexity under few-shot conditions, while HybridSN (OA: 94.67%), A2S2K (OA: 93.95%), and the contrastive learning method HiBiCo (OA: 94.69%) all clustered in the 94%–95% performance range. This result confirms an important trend: although these mainstream methods are effective, their performance seems to have hit a bottleneck on the Pavia dataset. However, GCN-based methods collectively surpassed this bottleneck, fully demonstrating the powerful ability of graph structure modeling to extract spatial contextual information. Among

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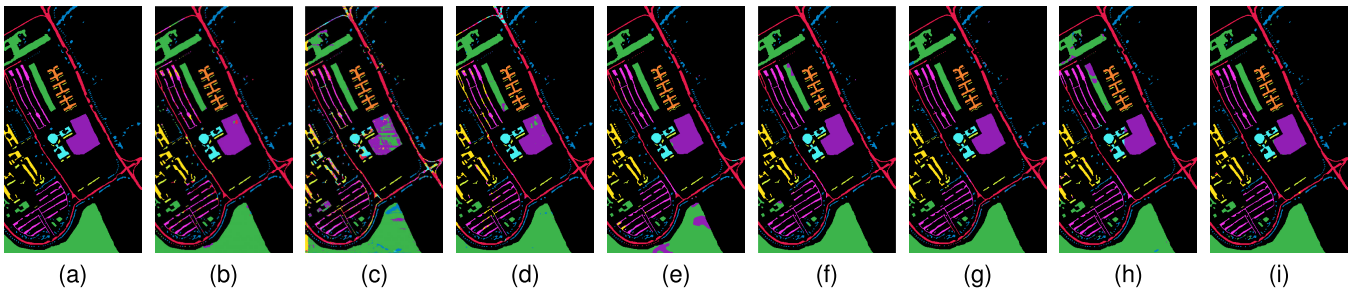


Fig. 11. Classification maps of different methods on the UP dataset. (a) Ground truth. (b) HybridSN. (c) RSSAN. (d) A2S2K. (e) HiBiCo. (f) AMGCFN. (g) GiGCN. (h) NESSGGCN. (i) CACL.

TABLE V
ABLATION STUDY ON MODULE CONTRIBUTIONS. BEST RESULTS ARE IN BOLDFACE

Case	CACL	SFEM	AMGCM	Metric	WHU	IP	UP
1	✓	✗	✗	OA	93.93±1.03	95.66±0.40	97.10±1.30
				AA	93.23±0.54	98.08±0.30	98.46±0.47
				Kappa	92.92±1.19	95.03±0.46	96.18±1.69
2	✓	✓	✗	OA	94.29±0.60	96.07±0.58	97.23±0.67
				AA	93.27±0.81	98.09±0.21	98.54±0.26
				Kappa	93.32±0.70	95.51±0.67	96.35±0.88
3	✓	✗	✓	OA	94.76±0.58	96.30±0.39	98.73±0.30
				AA	94.26±0.20	98.09±0.43	98.93±0.29
				Kappa	93.88±0.67	95.76±0.45	98.32±0.40
4	✓	✓	✓	OA	**95.29±0.46**	**96.73±0.50**	**99.17±0.24**
				AA	**94.54±0.45**	**98.25±0.29**	**99.29±0.16**
				Kappa	**94.49±0.54**	**96.25±0.57**	**98.91±0.31**

them, AMGCFN (OA: 98.05%) and GiGCN (OA: 97.99%) performed particularly well. Our method further achieved a performance leap, increasing the OA to 99.17% and the Kappa coefficient to 98.91%. This further indicates that the deep fusion of CNN and GCN features achieved through CACL can effectively enhance the model's ability to identify subtle feature differences in urban scenes, resulting in near-perfect classification.

The qualitative classification results on the UP dataset visually demonstrate the near-perfect classification capability of the proposed method. As shown in Fig. 11, compared to the high-performing GCN baseline model, the classification maps generated by our method exhibit significant improvements in both overall purity and fine-scale accuracy. Specifically, the approach almost completely eliminates minor noise within various land cover types and blurred inter-class boundaries. This visual enhancement aligns closely with the quantitative results, where our method achieved approximately a 1.1 percentage point increase in OA over the best GCN baseline. These findings strongly validate the critical role of CACL in extracting deep discriminative features, enabling the final classification results to approach a near-error-free state.

In summary, experiments on three representative hyperspectral datasets consistently demonstrate that our method achieves significantly superior performance compared to existing methods under extremely limited training samples. These results validate that the proposed CACL framework possesses strong generalization and robustness, making it effectively applicable to HSI classification tasks in different scenarios such as agriculture, woodland, and urban areas.

D. Ablation Study on Module Contributions

To systematically evaluate the effectiveness of the proposed SFEM and AMGCM, we designed four ablation experiments: 1) *Base Model:* Containing only the core mechanism of CACL; 2) *Base Model + SFEM:* Introducing the SFEM into the CNN branch of the base model; 3) *Base Model + AMGCM:* Introducing the AMGCM into the GCN branch of the Base Model; and 4) *Full Model:* The complete model integrating both SFEM and AMGCM. The results are shown in Table V.

First, compared to the Base Model, adding either new module individually brought stable performance improvements. After introducing SFEM, OA increased by 0.36%, 0.41%, and 0.13% on the WHU, IP, and UP datasets, respectively, validating the module's effectiveness in enhancing spectral pattern discriminability through frequency-domain analysis and alleviating spectral ambiguity under few-shot conditions. Introducing AMGCM resulted in more pronounced improvements, with OA increasing by 0.83%, 0.64%, and 1.63% on the WHU, IP, and UP datasets, respectively, proving that its adaptive multiscale graph construction mechanism can mine richer and more robust spatial hierarchical contextual information from scarce samples, strengthening the global feature representation.

Crucially, when both SFEM and AMGCM were integrated to form the Full Model, all evaluation metrics achieved the largest improvements. Compared to the Base Model, OA increased by 1.36%, 1.07%, and 2.07% on the WHU, IP, and UP datasets, respectively. This performance gain exceeds the simple sum of the improvements brought by

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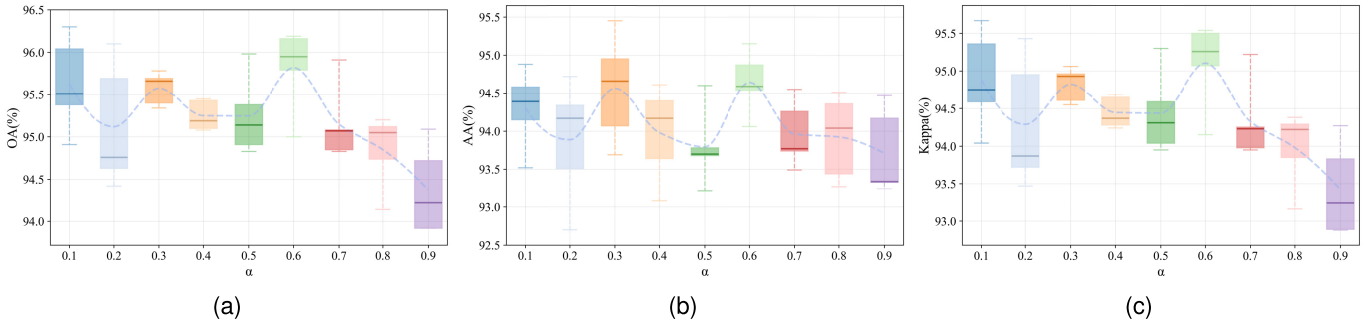


Fig. 12. Classification performance under different parameter α conditions on the WHU dataset. (a) OA. (b) AA. (c) Kappa.

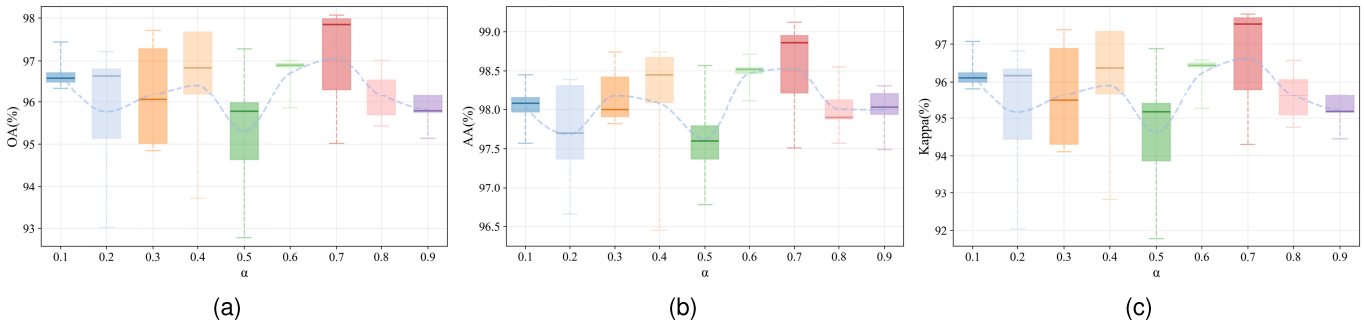


Fig. 13. Classification performance under different parameter α conditions on the IP dataset. (a) OA. (b) AA. (c) Kappa.

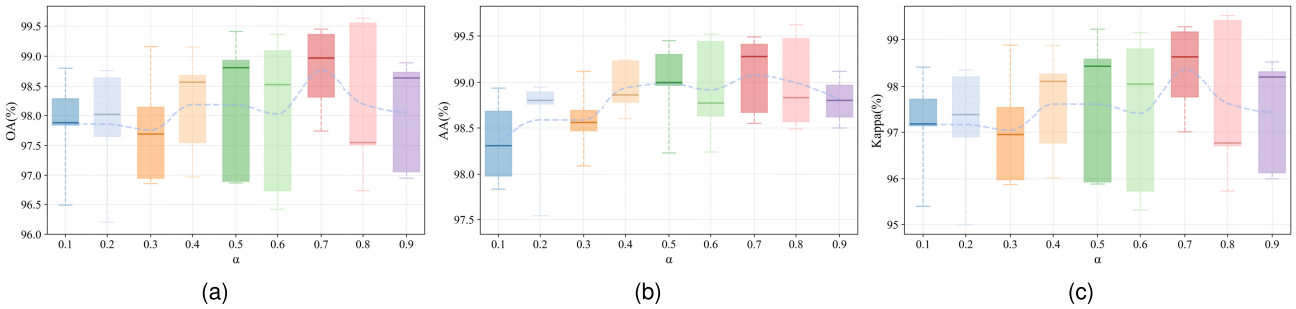


Fig. 14. Classification performance under different parameter α conditions on the UP dataset. (a) OA. (b) AA. (c) Kappa.

each module individually. This result strongly demonstrates a significant synergistic effect between the two modules. It indicates that the refined local spectral features provided by SFEM and the multiscale global spatial features captured by AMGCM achieve deep complementarity and fusion through the CACL mechanism, enabling the model to characterize the spatial–spectral characteristics of complex land cover more comprehensively and accurately, thus achieving optimal classification performance under the highly challenging few-shot condition.

E. Key Parameter Analysis

To systematically evaluate the impact of the fusion weight α for frequency-enhanced spectral features $\mathbf{F}_{\text{freq}}$ and spatial features $\mathbf{F}_{\text{spa}}$ on classification performance, we conducted detailed ablation experiments on three datasets. The value of α ranged from 0.1 to 0.9 with a step size of 0.1, and the experimental results are shown in Figs. 12–14. By analyzing the trend of classification performance with respect to α, it can be observed that, on the WHU dataset, the performance reached its global optimum at $\alpha = 0.6$. This dataset typically exhibits high spatial homogeneity (e.g., large and uniform crop fields), resulting in relatively low dependence on spatial context. Conversely, subtle spectral differences between crop categories are crucial for distinguishing crop types, making the frequency-enhanced spectral features significantly contributive. Hence, the optimal α value is relatively low. Both the IP and PaviaU datasets achieved optimal classification performance at $\alpha = 0.7$. These two datasets contain more complex land cover layouts and spatial details (e.g., fragmented field boundaries and dense urban building structures). Therefore, the model relies more on spatial features to capture local texture and structural information, leading to a higher optimal α value, meaning spatial features play a more dominant role. This difference directly reflects the inherent disparities among different data sources in terms of spatial structural complexity and interclass spectral separability. Overall, when α is in the range of 0.6–0.7, all three datasets achieve optimal or near-optimal classification performance. This indicates that a fusion strategy that assigns higher weight to spatial features while retaining a

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TABLE VI
MODEL COMPLEXITY COMPARISON ON THE IP DATASET

Method	Params (K)	FLOPs (M)
HybridSN	5122.42	499.12
RSSAN	254.85	28.05
A2S2K	370.86	171.60
HiBiCo	230.40	18.66
AMGCFN	368.22	2070.00
GiGCN	68.82	555.87
NESSGGCN	377.40	2001.20
CACL	391.84	2391.00

moderate but critical contribution (30%–40%) from frequency-domain features exhibits strong universality and robustness.

Regarding the branch balancing weight, we adopted $\delta = 0.4$. This specific setting was empirically validated in [41] to effectively balance the local representation from the CNN branch and the global topological information from the GCN branch. Adopting this value ensures that neither modality dominates, providing robustness for HSI datasets with complex spatial distributions.

F. Analysis of Computational Complexity

Table VI compares model complexity on the IP dataset. Regarding model parameters, the proposed CACL maintains a moderate scale, comparable to other mainstream GCN-based models such as AMGCFN and NESSGGCN. Specifically, GiGCN achieves the lowest parameter count due to its lightweight graph convolution design, whereas HybridSN exhibits the highest complexity, attributed to its dense 3D-CNN architecture. A significant difference in FLOPs exists between GCN-based and CNN-based models. This discrepancy originates from their fundamental processing mechanisms: GCN models operate on global or superpixel-level graphs, whereas CNNs process local image patches. Therefore, comparing CACL with other GCN-based methods is more meaningful. As shown, CACL's FLOPs are slightly higher than those of AMGCFN and NESSGGCN. This increase results from its robust architecture, which includes spatial operations in the CNN branch, frequency-domain processing in the SFEM, and multiscale graph convolutions in the AMGCM. Nevertheless, given CACL's superior classification performance under few-shot conditions, this additional computational cost is a justified tradeoff.

IV. CONCLUSION

In this study, the challenge of scarce labeled data in few-shot HSI classification was addressed by proposing a novel CACL framework that enables deep feature collaboration between CNNs and GCNs. By constructing positive sample pairs from local pixel-level features and global graph-structural features and performing explicit alignment within a shared embedding space, the proposed approach effectively overcame the limitations of existing methods in jointly representing local details and global structures. To further enhance the quality of positive pairs, an SFEM and an AMGCM were designed to strengthen spectral discriminability and enrich spatial information. Extensive experiments on multiple benchmark datasets confirmed that the proposed method achieves state-of-the-art performance with superior generalization under few-shot conditions. In particular, on the WHU, IP, and UP datasets under extremely limited training samples, it achieved the highest overall accuracies of 95.29%, 96.73%, and 99.17%, respectively, while also attaining superior AA and Kappa coefficients. However, it is worth noting that the proposed method relies on the quality of initial superpixel segmentation and introduces additional preprocessing costs for graph construction. In our future work, we plan to explore end-to-end adaptive graph learning strategies to mitigate these dependencies. Overall, this work provides an effective and generalizable solution for HSI classification with limited supervision and offers new insights into heterogeneous architecture fusion and contrastive representation learning in remote sensing.

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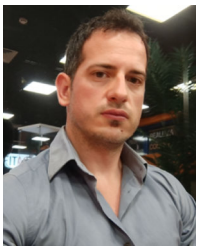
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